

6. (a) When a car stops the overall stopping distance is made up of two distances: the thinking distance and the braking distance. Increasing speed increases both the thinking distance and the braking distance.

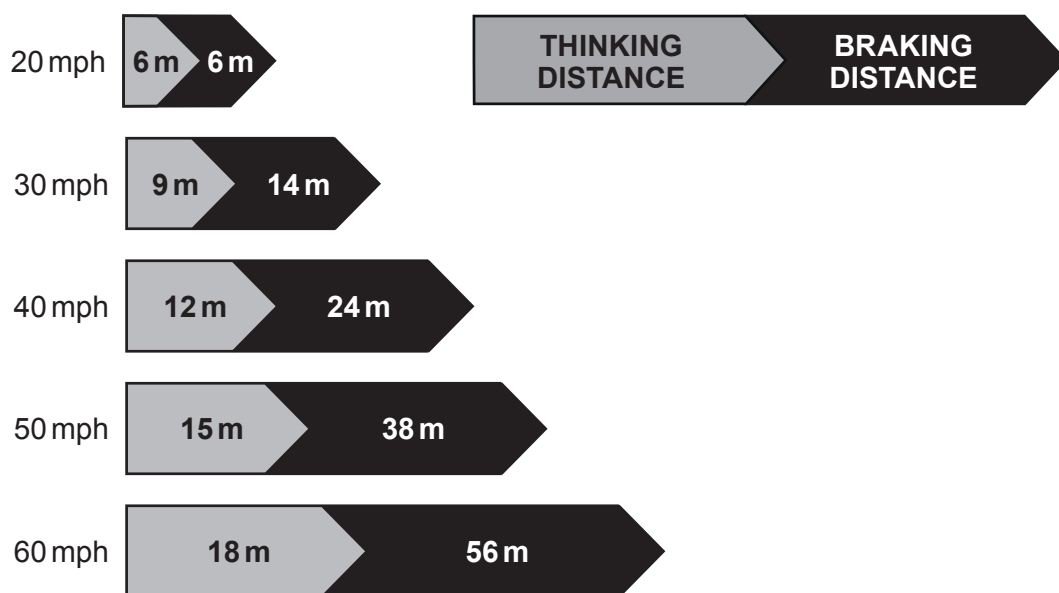
(i) State **one** factor, other than speed, which increases the thinking distance. [1]

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(ii) State **one** factor, other than speed, which increases the braking distance. [1]

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- (b) The diagram gives information about stopping distances at different speeds.



On a dangerous road, it is proposed to reduce the speed limit from 40 mph to 20 mph.

Bethan makes the following 3 suggestions.

1. The thinking distance will halve.
2. The braking distance will halve.
3. The overall stopping distance will halve.

Explain whether you agree with each suggestion.

Include data from the diagram to support your answer.

[3]

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